

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460959.0412 +/- 0.0091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.122 +/- 0.06
Transit depth (Rp/Rs)^2	1.49 +/- 1.46 [%]
Orbital Inclination (inc)	80.7 +/- 2.86
Airmass coefficient 1 (a1)	0.038 +/- 0.02
Airmass coefficient 2 (a2)	3.0 +/- 1.02
Scatter in the residuals of the lightcurve fit is	54.92 %
Best Comparison Star	None
Optimal Aperture	5.55
Optimal Annulus	12.95
Transit Duration (day)	0.007 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.122 ± 0.06 vs 0.1646 (deviation 0.45σ)
Duration fit vs published	0.007 d vs 0.0758 d (deviation 3.61σ)
Mid-time O–C	-2511.4 min
Depth SNR	1.3
Residual scatter	54.92 %

Light curve

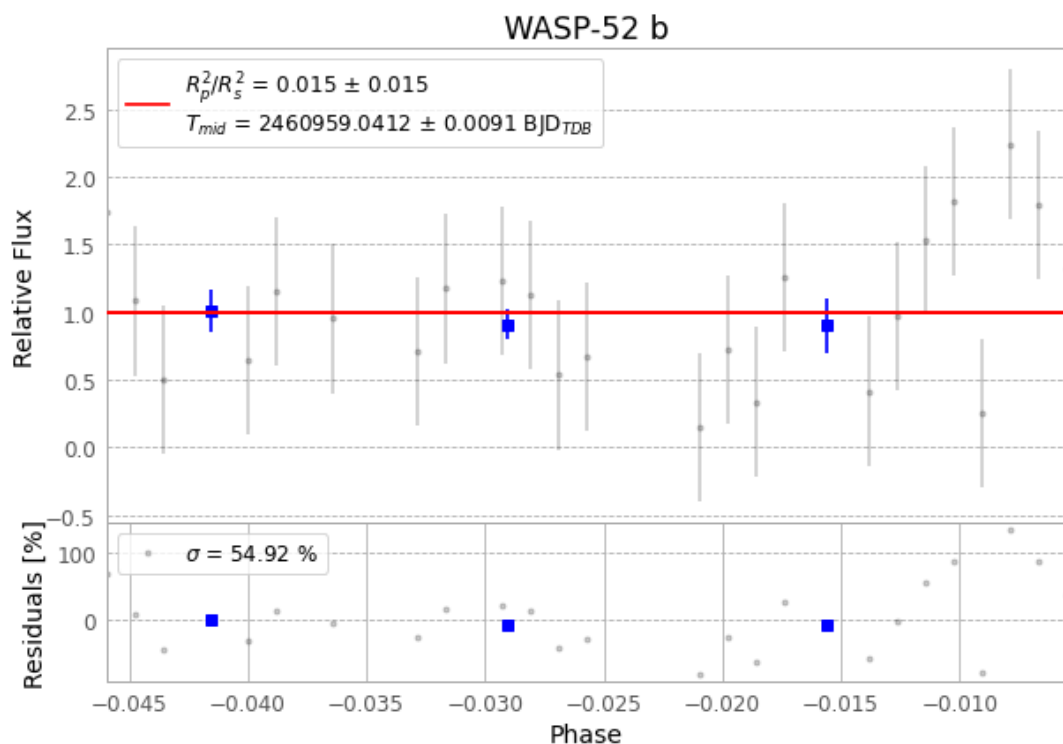


Figure 1 — FinalLightCurve_WASP-52 b_2025-10-11.png (SHA-256 9ef120089a9f7039...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a359777f0771211c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0

Data provenance

31 FITS frames (20 MB), WASP-52251012045415.FITS → WASP-52251012063615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-251012.log (182555445570...), FinalLightCurve_WASP-52 b_2025-10-11.png (9ef120089a9f...), FinalLightCurve_WASP-52 b_2025-10-11.csv (6ebb0efef800...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 17:30Z.